Mathematical Statistics

Lecture 3: Counting Probabilities with Combinatorics and the Factorial

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Contents

1 The Need for More Powerful Counting

In our last lecture, we defined the probability of an event A as:

$$P(A) = \frac{\text{Number of ways A can happen}}{\text{Total number of possible outcomes}}$$

This works perfectly for simple problems. However, what if we ask a more complex question like, "If I flip a coin 10 times, what is the probability of getting at least 5 heads?"

Manually listing and counting every single sequence is incredibly tedious and prone to error. The numbers get very large, very quickly. We need a more clever, systematic way to count. This is where **Combinatorics**, the mathematics of counting combinations and arrangements, becomes essential.

2 The Two Fundamental Questions of Counting

Before we can count anything, we must answer two critical questions about the scenario. The formula we use depends entirely on these answers.

1. **Does Order Matter?**
   - If **YES**, then the sequence '(A, B, C)' is different from '(C, B, A)'.
   - *Example:* A license plate 'KRB-57' is different from 'BRK-57'.
   - When order matters, we are counting **Permutations** or sequences.
2. **Is there Replacement?**
   - If **YES**, an item can be chosen more than once.
   - *Example:* In a coin flip, getting "Heads" on the first flip doesn't prevent you from getting "Heads" on the second.
   - If **NO**, once an item is chosen, it is removed from the pool of possibilities.
   - *Example:* When dealing cards from a deck, once the Ace of Spades is dealt, it cannot be dealt again.

3 An Essential Tool: The Factorial

The factorial is a mathematical operation that is a fundamental building block for combinatorics.

◎ Key Idea: The Factorial

Explanation: The factorial of a non-negative integer 'n', denoted by 'n¡, is the product of all positive integers less than or equal to 'n'.
The Formula:

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$$

By definition, $0! = 1$.
Example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

4 Counting Formulas and Worked Examples

Let's apply these concepts to derive formulas for different scenarios. We'll focus on counting the total number of outcomes (the denominator of our probability formula).

4.1 Scenario 1: Order Matters, With Replacement

Problem: How many unique sequences of 10 coin flips are possible?

- **Analysis:** Order matters ('HT' $\neq$ 'TH') and there is replacement (you can get Heads multiple times).
- **Solution:** For each of the 10 flips, there are 2 possibilities. The total is:

$$2 \times 2 \times \cdots \times 2 \text{ (10 times)} = 2^{10} = 1024$$

- **General Formula:** For selecting 'r' items from a pool of 'n' choices: Total Outcomes $= n^r$.

4.2 Scenario 2: Order Matters, Without Replacement (Permutations)

Problem: How many unique 5-card *runs* can be dealt from the top of a standard 52-card deck?

- **Analysis:** Order matters (it's a "run") and there is no replacement (a card can't be dealt twice).
- **Solution:**

$$52 \times 51 \times 50 \times 49 \times 48 = 311,875,200$$

- **General Formula (Permutations):** The number of permutations of choosing 'r' items from 'n' is:

$$P(n,r) = \frac{n!}{(n-r)!}$$

  For our example: $P(52,5) = \frac{52!}{(52-5)!} = \frac{52!}{47!} = 52 \times 51 \times 50 \times 49 \times 48$.

4.3 Scenario 3: Order Doesn't Matter, Without Replacement (Combinations)

Problem: How many unique 5-card poker *hands* are possible from a 52-card deck?

- **Analysis:** Order does **not** matter (a hand is a set of cards) and there is no replacement.
- **Logic:** We start with the number of permutations, $P(52,5)$, but this overcounts. For any set of 5 cards, there are $5!$ ways to arrange them. We must divide by this to correct the overcounting.

$$\text{Total Hands} = \frac{P(52,5)}{5!} = \frac{52!}{47! \times 5!} = 2,598,960$$

- **General Formula (Combinations):** This is read as "**n choose r**".

$$\binom{n}{r} = C(n,r) = \frac{n!}{r!(n-r)!}$$

5 Application: The License Plate Problem

Problem: A license plate consists of 4 letters (A-Z) followed by 2 numbers (0-9). Assume letters and numbers can be repeated. How many unique license plates are possible?

- **Analysis:** Order matters and there is replacement.
- **Letter combinations:** Choosing 4 from 26 with replacement: $26^4 = 456,976$.
- **Number combinations:** Choosing 2 from 10 with replacement: $10^2 = 100$.
- **Total Plates:** We multiply the possibilities for each part.

$$\text{Total} = 26^4 \times 10^2 = 45,697,600$$

Exercises

1. **Factorial Calculation:** Calculate the value of $\frac{8!}{5!}$.
2. **Combination Lock:** A suitcase has a combination lock with 3 dials, each with digits from 0 to 9. How many different combinations are possible? Which counting scenario does this represent?
3. **Electing Officers:** A club has 10 members. They need to elect a President, a Vice President, and a Treasurer. Assuming no person can hold more than one office, how many different ways can these three positions be filled? Which counting scenario does this represent?
4. **Forming a Committee:** The same club of 10 members decides to form a 3-person committee to plan an event. How many different committees can be formed? How does this problem differ from the one above?
5. **The Lottery:** A national lottery requires players to pick 6 different numbers from 1 to 49.
   (a) How many unique lottery tickets are possible?
   (b) If you buy one ticket, what is your probability of winning the grand prize (matching all 6 numbers)?

Solutions

1 We expand the factorials and cancel the common terms:

$$\frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} = \frac{8 \times 7 \times 6 \times 5!}{5!} = 8 \times 7 \times 6 = 336$$

2 This is **Scenario 1: Order matters, with replacement**.

- **Order matters** because the combination '1-2-3' is different from '3-2-1'.
- **With replacement** because a digit can be reused (e.g., '5-5-5' is a valid combination).

We have $n = 10$ choices (digits 0-9) and we are choosing $r = 3$ of them.

$$\text{Total Combinations} = n^r = 10^3 = 1000$$

3 This is **Scenario 2: Order matters, without replacement (Permutations)**.

- **Order matters** because the roles are distinct. (Person A as President, B as VP) is different from (Person B as President, A as VP).
- **Without replacement** because a person cannot hold more than one office.

We are choosing $r = 3$ people from $n = 10$.

$$P(10, 3) = \frac{10!}{(10-3)!} = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$$

There are 720 different ways to elect the officers.

4 This is **Scenario 3: Order doesn't matter, without replacement (Combinations)**.

- **Order does not matter** because all committee members have the same role. A committee of {Alice, Bob, Carol} is the same as {Bob, Carol, Alice}.
- **Without replacement** because you can't choose the same person twice for the committee.

We are choosing $r = 3$ people from $n = 10$.

$$\binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10!}{3!7!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = \frac{720}{6} = 120$$

There are 120 possible committees. This number is much smaller than the result in Problem 3 because we are not counting the different arrangements of the same three people. 5 This is a **Combination** problem because the order in which the numbers are drawn does not matter, and a number cannot be drawn more than once.

(a) How many unique lottery tickets are possible? We need to choose $r = 6$ numbers from a pool of $n = 49$.

$$\binom{49}{6} = \frac{49!}{6!(49-6)!} = \frac{49!}{6!43!} = \frac{49 \times 48 \times 47 \times 46 \times 45 \times 44}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 13,983,816$$

There are nearly 14 million possible tickets.

(b) What is your probability of winning? There is only one winning combination. The total number of possible outcomes is the number we just calculated.

$$P(\text{Win}) = \frac{\text{Number of winning tickets}}{\text{Total possible tickets}} = \frac{1}{13,983,816}$$

6 Summary of Lecture 3

Lecture Summary

1. Simple enumeration fails for large problems, requiring the systematic counting methods of **Combinatorics**.
2. All counting problems begin by asking two questions: **Does order matter?** and **Is there replacement?**
3. The **Factorial** ($n!$) is a crucial tool for calculating permutations and combinations.
4. We learned three key counting formulas, summarized below:

Scenario	Formula	Common Name
Order Matters, With Replacement	n^r	Power Rule
Order Matters, Without Replacement	$P(n,r) = \frac{n!}{(n-r)!}$	Permutations
Order Doesn't Matter, Without Replacement	$\binom{n}{r} = \frac{n!}{r!(n-r)!}$	Combinations

7 Why We Need Formalism

In previous lectures, we used counting to find probabilities for simple events. As problems become more complex, we need a formal mathematical language to define our terms precisely. **Set Theory** provides this language. It gives us a robust framework to define the space of possibilities and the specific events we care about, allowing us to build more general and powerful rules.

8 The Sample Space and Events

8.1 The Sample Space (Ω)

◎ Key Idea: The Sample Space (Ω)

The **Sample Space**, denoted by the Greek letter Omega (Ω), is the **set of all possible outcomes** of a random experiment. It's a complete list of everything that could possibly happen.

- **Experiment:** An action with an uncertain outcome (e.g., flipping 3 coins, rolling two dice).
- **Outcome (ω):** A single result from an experiment (e.g., getting 'Heads-Tails-Heads'). An outcome is an element of the sample space, written as $\omega \in \Omega$.

8.2 Events as Subsets

◎ Key Idea: An Event (A)

An **Event**, typically denoted by a capital letter like A or B, is a specific outcome or a collection of outcomes that we are interested in. Formally, an event is a **subset of the sample space** (Ω). We write this as $A \subset \Omega$.

If we conduct an experiment and the result is an outcome ω that is contained within the set A (i.e., $\omega \in A$), we say that "event A has occurred."

9 Worked Example: Three Coin Flips

Experiment: Flip a fair coin 3 times.

Step 1: Define the Sample Space (Ω)
First, we list every single possible outcome.

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

The total number of possible outcomes is $2^3 = 8$.

Step 2: Define Events as Subsets

- **Event A: "The first flip is a head."**

$$A = \{HHH, HHT, HTH, HTT\}$$

- **Event B: "The second flip is a tail."**

$$B = \{HTH, HTT, TTH, TTT\}$$

10 Operations on Events (Sets)

We can combine and modify events using standard set operations, which are often visualized with **Venn Diagrams**. Imagine a rectangle representing the entire sample space Ω, with circles inside representing events A and B.

10.1 Union (OR)

- **Notation:** $A \cup B$
- **Meaning:** The event that "A **OR** B (or both) occurs."
- **Example (3 Coin Flips):** $A \cup B = \{HHH, HHT, HTH, HTT, TTH, TTT\}$

10.2 Intersection (AND)

- **Notation:** $A \cap B$
- **Meaning:** The event that "A **AND** B both occur."
- **Example:** $A \cap B = \{HTH, HTT\}$

10.3 Complement (NOT)

- **Notation:** A^c or A'
- **Meaning:** The event that "A does **NOT** occur."
- **Example:** $A^c = \{THH, THT, TTH, TTT\}$ (the first flip is a tail).

11 The Axioms of Probability

To handle general cases (like a biased coin), we need a formal definition of probability. A probability measure, P, is a function that assigns a number between 0 and 1 to every event. It MUST obey three fundamental rules, called the **Axioms of Probability**.

Axiom 1: Non-negativity For any event A, $P(A) \geq 0$.

Axiom 2: Total Probability The probability of the entire sample space is 1, so $P(\Omega) = 1$.

Axiom 3: Additivity for Disjoint Events If two events A and B are **disjoint** (they have no outcomes in common, i.e., $A \cap B = \emptyset$), then the probability of A or B occurring is the sum of their individual probabilities:

$$\text{If } A \cap B = \emptyset, \text{ then } P(A \cup B) = P(A) + P(B)$$

12 Important Properties Derived from the Axioms

From these three simple axioms, we can derive many other useful properties of probability.

1. **Probability of the Complement:** $P(A^c) = 1 - P(A)$
2. **Probability of the Impossible Event:** $P(\emptyset) = 0$

3. **The General Addition Rule (Inclusion-Exclusion Principle):** For *any* two events A and B (even if they are not disjoint):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Lecture Summary

1. **Set Theory** provides the formal language for modern probability.
2. The **Sample Space (Ω)** is the set of all possible outcomes.
3. An **Event (A)** is a subset of the sample space ($A \subset \Omega$).
4. We can combine events using **Union (OR)**, **Intersection (AND)**, and **Complement (NOT)**.
5. A valid probability measure must obey the three **Axioms of Probability**.
6. These axioms allow us to derive essential rules, like the formula for complements and the general addition rule.

Exercises

1. **Rolling a Die:** Consider the experiment of rolling a single, fair six-sided die.
   (a) Define the sample space Ω.
   (b) Let A be the event that the outcome is an even number. List the elements of A.
   (c) Let B be the event that the outcome is greater than 4. List the elements of B.
2. **Set Operations:** Using the events A and B from Problem 1:
   (a) Find the union, $A \cup B$.
   (b) Find the intersection, $A \cap B$.
   (c) Find the complement of A, A^c.
   (d) Calculate $P(A)$, $P(B)$, $P(A \cup B)$, and $P(A \cap B)$.
3. **Applying the Addition Rule:** A survey finds that 60% of people own a car (Event C), 30% own a motorcycle (Event M), and 15% own both. What is the probability that a randomly selected person owns a car OR a motorcycle?
4. **Applying the Complement Rule:** If the probability of rain tomorrow is 0.4, what is the probability that it will not rain?
5. **Disjoint Events:** If events A and B are disjoint, what is the value of $P(A \cap B)$?

Solutions

1 **(a) Sample Space:** The set of all possible outcomes is $\Omega = \{1, 2, 3, 4, 5, 6\}$.
(b) Event A (even number): The outcomes that are even are $A = \{2, 4, 6\}$.
(c) Event B (greater than 4): The outcomes greater than 4 are $B = \{5, 6\}$.
2 **(a) Union** $A \cup B$**:** The set of outcomes in A OR B. $A \cup B = \{2, 4, 5, 6\}$.
(b) Intersection $A \cap B$**:** The set of outcomes in A AND B. The only common element is 6. $A \cap B = \{6\}$.
(c) Complement A^c**:** Everything in Ω that is NOT in A. $A^c = \{1, 3, 5\}$.
(d) Probabilities: Since the die is fair, each outcome has probability 1/6.

- $P(A) = \frac{\text{Number of elements in A}}{|\Omega|} = \frac{3}{6} = 0.5$
- $P(B) = \frac{\text{Number of elements in B}}{|\Omega|} = \frac{2}{6} = \frac{1}{3} \approx 0.333$
- $P(A \cup B) = \frac{\text{Number of elements in } A \cup B}{|\Omega|} = \frac{4}{6} = \frac{2}{3} \approx 0.667$
- $P(A \cap B) = \frac{\text{Number of elements in } A \cap B}{|\Omega|} = \frac{1}{6} \approx 0.167$

We can verify the addition rule: $P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6}$, which matches $P(A \cup B)$.
3 We are given: $P(C) = 0.60$, $P(M) = 0.30$, and $P(C \cap M) = 0.15$. We want to find $P(C \cup M)$.
Using the General Addition Rule:

$$P(C \cup M) = P(C) + P(M) - P(C \cap M)$$

$$P(C \cup M) = 0.60 + 0.30 - 0.15 = 0.75$$

The probability that a person owns a car or a motorcycle is 75%.

4 Let R be the event that it rains. We are given $P(R) = 0.4$. We want to find the probability that it does not rain, which is the complement, $P(R^c)$.

Using the Complement Rule:

$$P(R^c) = 1 - P(R) = 1 - 0.4 = 0.6$$

The probability that it will not rain is 0.6 or 60%.

5 By definition, two events A and B are disjoint if they have no outcomes in common. This means their intersection is the empty set: $A \cap B = \emptyset$.

The probability of the empty set (an impossible event) is always zero. Therefore:

$$P(A \cap B) = P(\emptyset) = 0$$